# COVID-19: Clinical aid VERSION 6.0 (Jan 2021)

## Danger *Aims Asses PPE requirements*

| ***Checks*** | ***Treatment protocol*** |
|---|---|
| Check Covid PCR | Stop PPE at 14 days if: PCR -ve, T<37.8°C for >48hrs **AND** clinical improving (inc resp) |
| | If negative; but clinically likely COVID, continue enhanced PPE, D/W Micro + Infec.Control |

## Airway *Aims Safe airway*

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|---|---|
| Check tube length or leak | Discuss with urgently with airway doctor |
| > 10-14 days ventilated? | Expert advice for trachostomy - infection control vs weaning |

## Breathing *Aims pH > 7.20 Sats 90% $pO_2$ 8 6-8ml/kg ideal body weight TV*

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|---|---|
| Ventilator safe? | 6-8ml / kg tidal volume IBW; min PEEP 5-8 (Aim <10 if Compliance <40) |
| | Adequate assist/ASB |
| Oxygen ≤ 40%? | Wean to assist/CPAP-ASB |
| | Stable for 12hrs -> Leak test + expert advice -> extubation to facemask/CPAP |
| Oxygen 40% - 60%? | Check Sputum plugging/ pneumothorax/ secondary infection/ tube position |
| | Consider CXR / Adjust PEEP / Recruitment manoeuvres / Echo |
| Oxygen ≥60%? Do above plus: | Higher PEEP trial 10 - 12 - 14 – 16 (+/- echo, risk of acute cor pulmonale) |
| | Use atracurium infusion for paralysis |
| | No improvement – **prone early if P/F <20 for 16-18hrs/24hrs** |
| | No improvement - consider APRV then discuss ECMO referral with ICM team. |

## Circulation *Aims MAP 65-70 Neutral or negative fluid balance*

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|---|---|
| Norad > 0.5 mcg/kg/min? (20mls/h at 8mg/50mls) | Consider ECHO or CO monitor |
| | Consider hydrocortisone 200mg/24hrs and hold diuretics |
| | Consider second vasopressor/inotrope |
| | Consider infection, see below |
| Positive fluid balance? | Add furosemide 20mg BD or increase current dose. Consider Echo. |
| | Consider dialysis if oxygen > 60% |
| CPR and escalation decisions? | Family & expert advice & national guidance |
| Cardiomyopathy (potentially lethal late on) | Avoid excessive inotrope use: MAP 65-70<br>Check Cardiac function before wakening pref. ECHO |

## Sedation *Aims Calm and safe if stable Deeply sedated if unstable*

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|---|---|
| If $FiO_2$ ≤ 40%? (N.B check for cardiomyopathy as above) | Daily sedation hold or stop long acting agents, especially midazolam |
| | Wean sedation. Consider clonidine infusion if agitated and high BP |

## Exposure Aims *Minimum number of procedures and lines*

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|---|---|
| Central line? | Remove/switch if signs of CVP infection / not needed |

## Food and family *Aims Fed Opening bowels every day Family updated*

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|---|---|
| Feed commenced? | NG feed + prokinetics if aspirate >500ml (>300ml if proned, seek Dietician advice) |
| | Insulin / glucose if needed |
| Bowels opened? | All patients receive Lactulose 20ml BD plus Senna 15mls OD unless diarrhoea |
| Family updated? | Phone/Skype/dedicated family update team |

## Heamatology *Aims Review and act on all results*

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|---|---|
| Blood tests? | Daily COVID set (inc Trop, Ferritin, D-Dimer) + Procalcitonin if deciding to start or stop ABx |
| Thromboprophylaxis: | Enoxaparin: If <50Kg 20mg BD, 50-100Kg 40mg BD, 100-150Kg 60mg BD. If Wt >150Kg, CrCl <30 or high risk bleeding see BCU guideline. Reduce to OD at critical care discharge unless felt to be at high risk. |

## Infection/drugs *Aim Treat Covid, Get a diagnosis, Review/minimise all drugs*

| | |
|---|---|
| COVID Drugs | Consider second dose of Tocilizumab within 12-24 hours of first dose if insufficient clinical improvement.<br>Dexamethasone 6.6mg OD for max 10 days<br>Remdesivir maintenance 100mg OD for total 5 days max. **STOP if eGFR <30 or RRT, ALT >5 x normal, significant deteriorating ALP/Bili/PT.** |
| Samples collected? | Dry viral swab from throat or sputum for COVID-19, respiratory viruses |
| | Sputum sample (fungal?), Urine for Legionella/Pneumococcus |
| | Carbocysteine 750mg NG TDS if thick sputum |
| | Omeprazole 40mg OD IV. When enterically fed, switch to Lansoprazole fastab 30mg OD NG |
| | FlowTrons or TEDs DVT prevention |
| Is there secondary infection? (If deteriorating, worsening infective markers or increased sputum) | Review Procalcitonin and review need for ABx– See separate guidance.<br>See Microguide for prescribing advice for CAP or VAP. |
| | If deteriorating, send serum Beta-D-Glucan, Galactomannan, NBAL or BAL for fungus + C&S. Discuss antifungal (usually caspofungin) with Micro. |